DATA STRUCTURES • MINI PROJECT REPORT

Attendance Management System

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1 Overview

This console program keeps attendance for a class. A teacher adds students, marks each day as present or absent, and the program works out every attendance percentage. Records can be searched, ranked and checked against a 75% limit, and are kept in memory while the program runs.

2 Data Structure Concepts Used

Dynamic Array

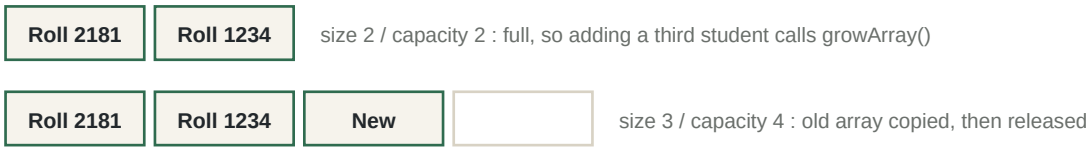
Records live in an array made with `new[]`. It starts at 2 slots and doubles when full; the old array is copied across and freed.

Searching

Binary search finds a roll number in the sorted array. Linear search finds students by name.

Sorting

Insertion sort orders students by roll number. Selection sort ranks them by attendance percentage, highest first.



An array suits this project: the class size is not fixed, and binary search and both sorts need access by index.

3 Program Functions

Function	What it does
`addStudent()`	Adds a student with 0/0 attendance, rejects duplicate roll numbers, calls growArray() when the array is full.
`growArray()`	Creates an array twice as big, copies the records across and frees the old array.
`markAttendance()`	Asks P or A for every student and updates attended and total.
`percentage()`	Returns attended × 100.0 ÷ total as a double; returns 0 if total is 0.
`binarySearch()`	Sorts by roll number, then searches for the roll number by halving the range.
`linearSearch()`	Compares the typed name with each stored name from first to last.
`sortByRoll()`	Insertion sort, ascending by roll number.
`sortByPercentage()`	Selection sort, highest percentage first.
`showDefaulters()`	Lists everyone below 75%.
`deleteStudent()`	Removes a student and shifts the later records left.
`readInt() readName() readPA()`	Safe input: roll number must be a whole number, name letters only, attendance P or A.

4 How the Algorithms Work

Algorithm	Steps
Binary search	Set low = 0 and high = size – 1. Look at the middle roll number; go right if it is smaller than the target, left if larger. Stop when it matches or low passes high.
Linear search	Start at the first student and compare the stored name with the typed name. Stop when they match, or report not found after the last student.
Insertion sort	For each student from index 1, shift larger roll numbers one place right and drop the student into the gap.
Selection sort	For each position, find the highest percentage in the unsorted part and swap it into place.
Delete	Find the index of the roll number, move each later record one place left, and decrease size by 1.

5 Time and Space Complexity

Operation	Time	Why
Add student	O(n)	Duplicate check scans the array; a resize occasionally copies n records.
Mark attendance / display	O(n)	Visits each student once.
Binary search	O(log n)	Range halves every step once the array is sorted by roll number.
Linear search (name)	O(n)	May check every student.
Insertion sort	$O(n^2)$	Worst case shifts every earlier record; O(n) when already sorted.
Selection sort	$O(n^2)$	Two nested loops in every case.
Delete student	O(n)	Locate the record, then shift the rest.
Space	O(n)	n students stored; searching and both sorts need only O(1) extra space.

6 Coding Problem and AI Use

Problem: When a name such as “Hasnain Qazi” was typed where a roll number was expected, the input went into an error state and the menu printed endlessly; names also accepted digits. **AI use:** AI helped make the roll number accept only whole numbers and the name only letters, using small input functions that check the value and ask again.

Field	Wrong input	Before the fix	After the fix
Roll number	Letters, e.g. Hasnain Qazi	Menu repeated endlessly	“Please enter a number.”, then asks again
Name	Digits or symbols, e.g. Ali123	Accepted	“Name must contain letters only.”, then asks again
Attendance	Anything except P or A	Wrong record saved	“Enter P or A only.”, then asks again

AI prompt used: “My C++ program gets stuck when I type letters in the roll number. Make the roll number accept only integers and the name accept only alphabets, and explain how it works.”

Percentage problem (project sheet): `(attended / total) * 100` divides two integers first, so 3 out of 4 gave 0%. `(attended * 100.0) / total` gives 75.00%.

7 Test Run (actual program output)

Tested with Roll 2181 Hasnain Qazi and Roll 1234 Khalilullah over two days of attendance.

```
> Add student: letters typed in the roll number
Roll number: Hasnain Qazi
Please enter a number.
Roll number: 2181
Name: Hasnain Qazi
Student added.
Roll number: 1234    Name: Khalilullah
Student added.
> Mark attendance, day 1 and day 2
2181 Hasnain Qazi (P/A): p    1234 Khalilullah (P/A): p
2181 Hasnain Qazi (P/A): p    1234 Khalilullah (P/A): a
> Display all
Roll: 2181 | Name: Hasnain Qazi | Attended: 2/2 | 100.00%
Roll: 1234 | Name: Khalilullah | Attended: 1/2 | 50.00%
> Show students below 75%
Roll: 1234 | Name: Khalilullah | Attended: 1/2 | 50.00%
> Delete roll 1234, then display all
Student deleted.
Roll: 2181 | Name: Hasnain Qazi | Attended: 2/2 | 100.00%
```

8 Conclusion

The project applies a dynamic array, searching and sorting together in one menu-driven C++ program. Every input is checked (whole-number roll, letters-only name, P or A attendance), duplicate roll numbers are rejected, and the attendance percentage is calculated correctly.